PORTFOLIO STRESS TESTING & SCENARIO ANALYSIS PROJECT

1. Introduction

Financial institutions face significant exposure to market risk due to fluctuations in asset prices. While traditional risk measures such as Value at Risk (VaR) provide useful estimates under normal market conditions, they often underestimate losses during extreme events—highlighted by the 2008 Global Financial Crisis.

To address these limitations, stress testing and scenario analysis are widely used to evaluate portfolio resilience under adverse conditions.

This project develops a comprehensive stress testing framework for a multi-asset portfolio using historical market data. It integrates:

- **Historical risk analysis**
- **Scenario-based stress testing**
- **Monte Carlo simulation**

The portfolio consists of four assets representing different market characteristics:

- **Apple Inc. (AAPL) – large-cap technology**
- **JPMorgan Chase & Co. (JPM) – banking sector**
- **Tesla Inc. (TSLA) – high-growth, high-volatility**
- **Vanguard FTSE Emerging Markets ETF (VWO) – emerging markets exposure**

The objective is to evaluate portfolio risk under both observed and simulated extreme conditions, using a fully reproducible pipeline implemented in R and SQL.

2. Data and Methodology

2.1 Data Extraction

Historical daily price data from January 2020 onward was sourced from Yahoo Finance using R. Data was stored in a MySQL database (financedata_db) to ensure:

- **Structured storage**
- **Efficient querying**
- **Reproducibility**

R was used for analysis, while SQL served as the data storage layer.

2.2 Data Processing

The dataset was structured in long format and transformed into a wide format for portfolio analysis.

Daily log returns were computed:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

This approach ensures:

- **time additivity**
- **statistical consistency**

Missing values from lag operations were removed to maintain data integrity.

2.3 Portfolio Construction

An equally weighted portfolio was constructed:

- **AAPL: 25%**
- **JPM: 25%**
- **TSLA: 25%**
- **VWO: 25%**

Portfolio returns were computed as:

$$R_p = \sum w_i R_i$$

This provides a simple yet effective framework for analyzing diversification effects.

3. Exploratory Data Analysis

3.1 Price Dynamics

Asset prices exhibited:

- **clear bull and bear cycles**
- **sharp drawdowns (market shocks)**
- **varying volatility across assets**

Tesla showed the highest volatility, while Apple exhibited relatively stable growth.

3.2 Return Distributions Key findings:

- **Mean returns ≈ 0**
- **Presence of fat tails**
- **Significant variation in dispersion**

This indicates a higher probability of extreme events than predicted by a normal distribution.

3.3 Correlation Structure

The correlation matrix showed:

- **moderate positive correlations**
- **no perfect co-movement**

This suggests diversification benefits under normal conditions, though these may weaken during stress.

4. Risk Measurement

4.1 Portfolio Behavior

The portfolio return series exhibited:

- **reduced volatility compared to individual assets**
- **persistence of market-driven fluctuations**

- **occasional extreme movements**

4.2 Stylized Facts

The analysis confirmed key financial characteristics:

- **Volatility clustering**
- **Fat tails**
- **Asymmetric risk behavior**

These violate standard model assumptions and justify advanced risk modeling.

4.3 Risk Metrics

The following were computed:

- **Mean return**
- **Standard deviation**
- **Value at Risk (VaR)**
- **Expected Shortfall (ES)**

Key insight:

Expected Shortfall consistently exceeds VaR, highlighting tail risk that VaR alone fails to capture.

5. Scenario Analysis

Three stress scenarios were applied:

5.1 Market Crash (-20%)

The market crash scenario produced the most severe deterioration in portfolio performance, with significantly amplified downside risk and a pronounced left-tail shift in the return distribution. This indicates that the portfolio is highly exposed to systemic equity risk, where diversification fails as correlations converge during periods of market stress.

5.2 Interest Rate Shock

The interest rate shock scenario revealed asymmetric sector sensitivity, with growth-oriented assets exhibiting greater volatility compared to financial institutions. This suggests that portfolio risk is not uniformly distributed across assets but is influenced by macroeconomic factors such as monetary policy.

5.3 Emerging Market Crisis

The emerging market crisis scenario demonstrated that localized shocks can propagate through the portfolio via spillover effects. While diversification partially mitigates losses, exposure to emerging markets introduces additional vulnerability to global financial instability.

5.4 Stress Comparison

- **Market crash → highest risk**
- **Rate shock → moderate**
- **EM crisis → contained**

Key insight:

Diversification weakens significantly during systemic shocks.

5.5 VaR and ES Under Stress

The consistent divergence between Value at Risk and Expected Shortfall highlights the limitations of VaR as a standalone risk measure. Expected Shortfall provides a more comprehensive view of extreme losses by capturing tail risk beyond the VaR threshold.

This confirms tail risk amplifies disproportionately during crises.

6. Monte Carlo Simulation

A Monte Carlo simulation (10,000 scenarios) was performed using a multivariate normal distribution based on historical mean and covariance.

While Monte Carlo simulation provides a forward-looking distribution of potential outcomes, its reliance on the normality assumption may lead to an underestimation of extreme losses. This reinforces the importance of complementing simulation with stress testing.

Key Findings:

- **Simulated distribution confirms tail risk**
- **Monte Carlo VaR aligns with historical estimates**
- **Differences highlight model assumptions**

Critical Insight:

The assumption of normality may underestimate extreme losses, reinforcing the importance of stress testing.

7. Discussion

7.1 Diversification

- **Effective under normal conditions**
- **Breaks down during stress**

7.2 Risk Drivers

Portfolio risk is driven by:

- **asset volatility**
- **correlation dynamics**
- **exposure to extreme events**

7.3 Key Findings

- **Risk is underestimated under normal assumptions**
- **Tail events significantly impact performance**
- **Stress testing provides critical additional insight**

8. Conclusion

This project demonstrates a fully integrated stress testing framework combining:

- **historical risk analysis**
- **scenario-based shocks**
- **Monte Carlo simulation**

The results show that:

- **portfolios are highly vulnerable to systemic shocks**
- **diversification has limits**
- **traditional models underestimate extreme risk**

Combining multiple approaches provides a more robust and realistic assessment of portfolio risk.

9. Limitations & Future Work

Limitations

- **Normality assumption (underestimates tail risk)**
- **Static portfolio weights**
- **Reliance on historical data**

Future Improvements

- **Fat-tailed distributions (t-distribution)**
- **Volatility modeling (GARCH)**
- **Dynamic portfolio optimization**
- **Macro-driven stress scenarios**